

SANKET KUMAR

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EDUCATION

Bachelor of Technology in Mechanical Engineering

Expected 2027

National Institute of Technology Srinagar

Current Final Year Student

CGPA: 7.78

EXPERIENCE

Winter Intern, Solid State Physics Laboratory (SSPL), DRDO, Delhi

Dec 2025 – Jan 2026

- Developed a numerical CFD model of the regenerator in a Stirling cryocooler using ANSYS Fluent to investigate its thermo-fluid performance under transient operating conditions.
- Simulated a porous wire-mesh regenerator operating with helium at a mean pressure of **30 bar** and an operating frequency of **50 Hz**.
- Modeled a regenerator with dimensions of **6.5 mm diameter** and **50 mm length** for a cryocooler providing a cooling capacity of **0.5 W at 80 K**, while maintaining the hot end at room temperature.
- Analyzed temperature distribution, pressure drop, heat transfer characteristics, and regenerator effectiveness to evaluate cryocooler performance and thermal efficiency.

PROJECTS

Design and Fabrication of Solar-Assisted Biogas Plant for Cold Climatic Conditions — B.Tech Final Year Project

- Fabricated a **100 L portable plug-flow anaerobic digester** with a **350 L gas holder**, designed in **SolidWorks** and insulated using **PUF** for efficient operation under Jammu & Kashmir winter conditions.
- Achieved an average biogas production of **16.5 L/day (82–95% CH₄)** from **2 kg/day** of food waste; integrated a **300 W solar panel**, **100 Ah battery**, and **200 W thermostatically controlled heater** to maintain **30–40°C**, enabling off-grid operation.
- Impact: Diverts approximately **730 kg** of organic waste annually, avoids nearly **1.2 t CO₂** emissions, and provides clean cooking fuel along with nutrient-rich **NPK biofertilizer** for rural households.

3D CAD Modeling of ABB Industrial Robot

- Created a complete 3D CAD model of an ABB industrial robot using SolidWorks.
- Modeled robot links, joints, and assemblies with appropriate geometric constraints.

SKILLS

Software	ANSYS , Hypermesh, SolidWorks, MATLAB, Microsoft Excel
Programming	Python (Basic) , C
Soft Skills	Leadership, Communication, Problem Solving, Teamwork, Time Management

EXTRACURRICULAR ACTIVITIES

Sponsorship Team Member

- Secured sponsorships and coordinated with industry partners for **Rang-e-Chinar 2024** and **Techvaganza 2025**, supporting event planning and stakeholder engagement.

Sports

- Qualified up to the **North Zone level** in **10 m Air Pistol Shooting**..